Week 1, Lecture 3
Quadratures and phase space

Lecturer: Josephine
Email: j.dias@uq.edu.au
Office: Parnell Room 320
Office hours: Thursdays 3-4 pm

Notes 1.2, 1.3, 2.2 and 2.3
Bachor & Ralph, Chapters 2.1, 4.2.2 and 4.4.1
Walls & Milburn, Chapter 2.3

Review

What you learned last time…

Single mode EM field → **Simple harmonic oscillator**

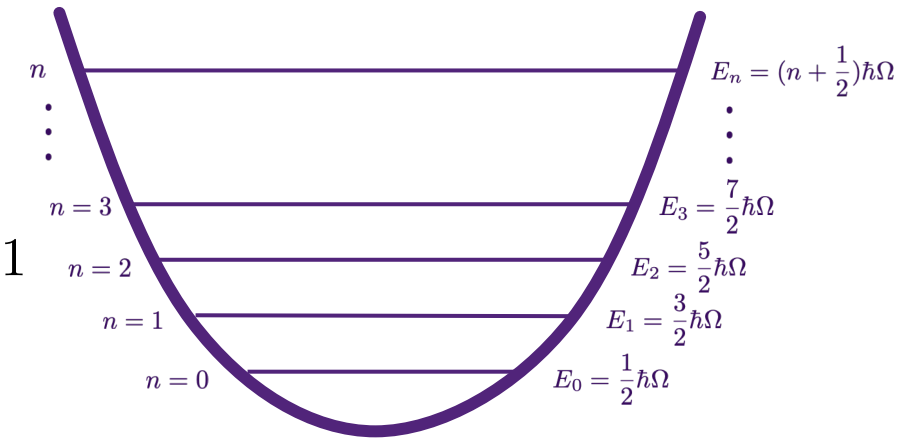
Hamiltonian: $\hat{H} = \hbar\Omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$

Eigenstates: $\{|n\rangle\} = \{|0\rangle, |1\rangle, |2\rangle, \cdots\}$

Annihilation and creation operators

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle \qquad [\hat{a}, \hat{a}^\dagger] = 1$$

$$\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$



Coherent states: brief review

A coherent state can be generated by displacing the vacuum:

$$|\alpha\rangle \equiv \hat{D}(\alpha)|0\rangle$$

where

$$\hat{D}(\alpha) = \exp\left(\alpha a^\dagger - \alpha^* a\right)$$

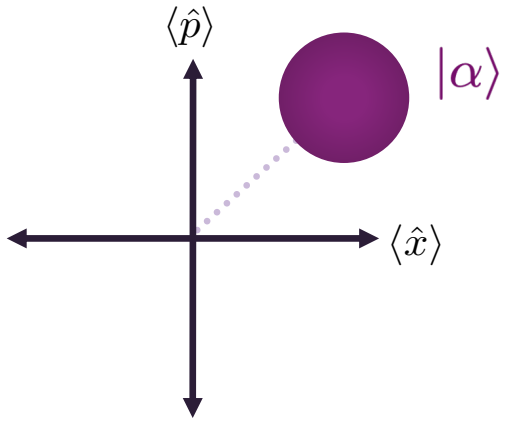
Coherent states are eigenstates of the annihilation operator:

$$a|\alpha\rangle = \alpha|\alpha\rangle$$

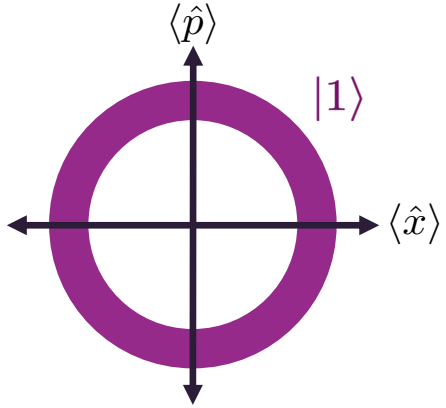
Here, α is the complex coherent amplitude.

In this lecture…

- Observables in quantum optics
- Quadrature amplitudes
- Describing quantum optical states in phase space



Coherent state



Number state

Observables in quantum mechanics

A physical observable is represented by a Hermitian operator:

$$\hat{A}^\dagger = \hat{A}$$

Position and momentum are observables:

$$\hat{x}^\dagger = \hat{x}, \qquad \hat{p}^\dagger = \hat{p}$$

Their expectation values are measurable average quantities:

$$\langle x \rangle = \langle \psi | \hat{x} | \psi \rangle, \qquad \langle p \rangle = \langle \psi | \hat{p} | \psi \rangle$$

Position and momentum do not commute:

$$[\hat{x}, \hat{p}] = i\hbar$$

Therefore,

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

Observables in quantum optics

Photon number $\hat{n} = \hat{a}^\dagger \hat{a}$

Define the dimensionless quadrature operators

$$X_1 = a^\dagger + a, \qquad X_2 = i\left(a^\dagger - a\right)$$

Both quadratures are Hermitian:

$$X_1^\dagger = X_1, \qquad X_2^\dagger = X_2$$

Their commutator is

$$[X_1, X_2] = 2i$$

Therefore,

$$\Delta X_1 \Delta X_2 \geq \frac{1}{2} \left|\langle [X_1, X_2] \rangle\right| = 1$$

In a future lecture, I will show you how we measure these quadratures

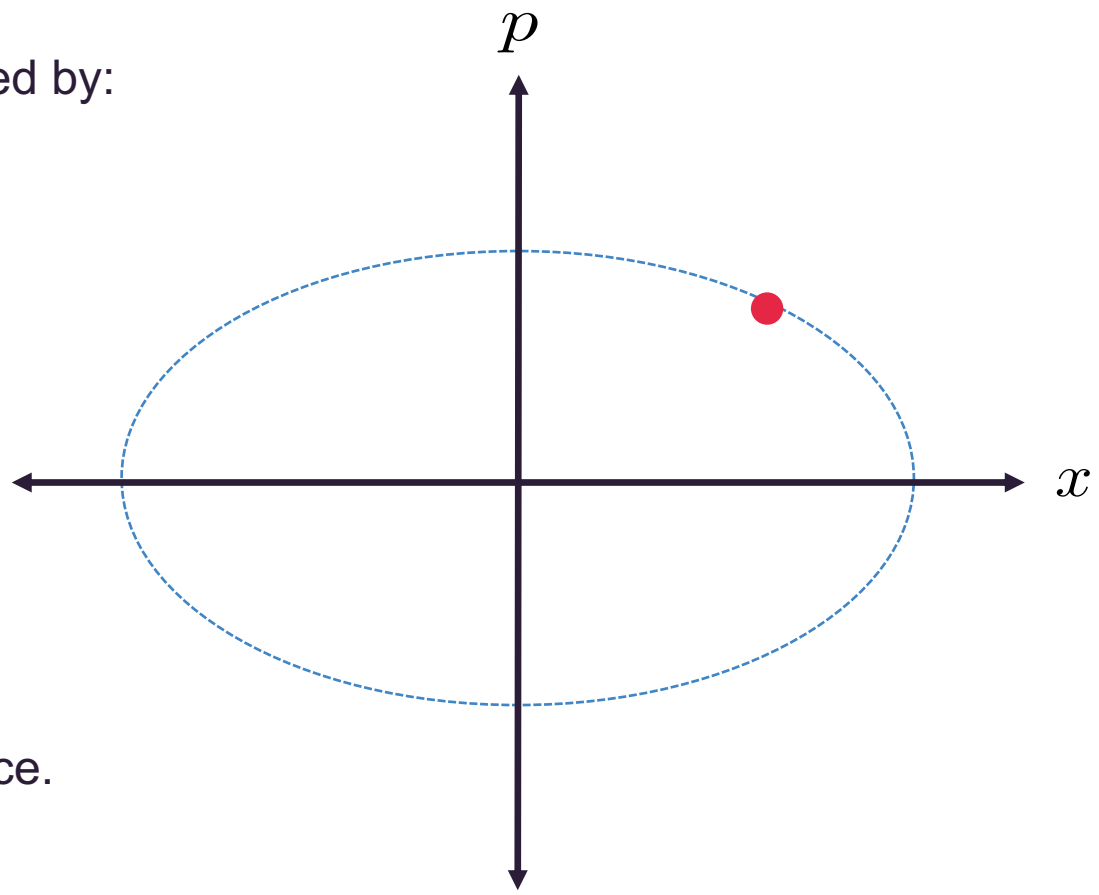
What is phase space?

Classical particles

The instantaneous state of a classical particle is specified by:

- its position x;
- its momentum p.

A point in phase space represents both quantities:



For a harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{1}{2}m\Omega^2 x^2$$

Constant-energy trajectories form ellipses in phase space.

Quantum phase space

Position and momentum are non-commuting observables:

$$[\hat{x}, \hat{p}] = i\hbar$$

Their uncertainties satisfy

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

For a quadrature $\hat{x}$,

Standard deviation $\Delta x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2}$

Variance $\mathrm{var}(x) = \Delta x^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$

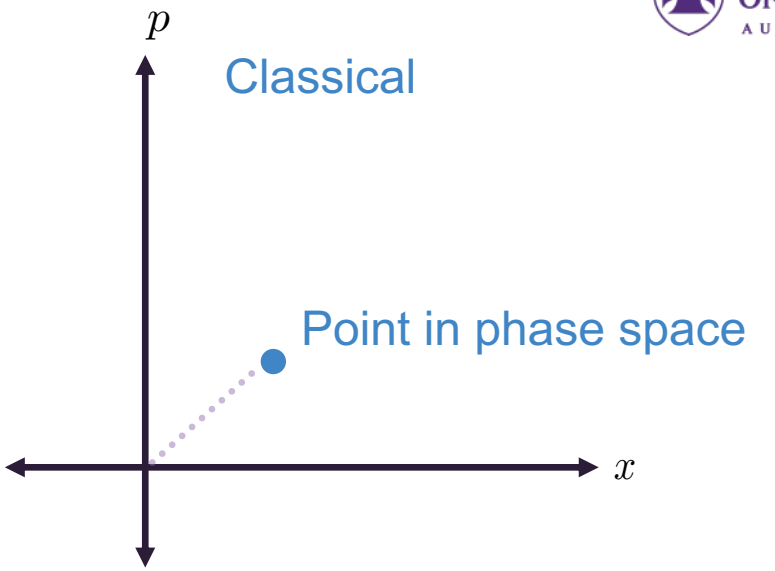
A quantum state is therefore described by:

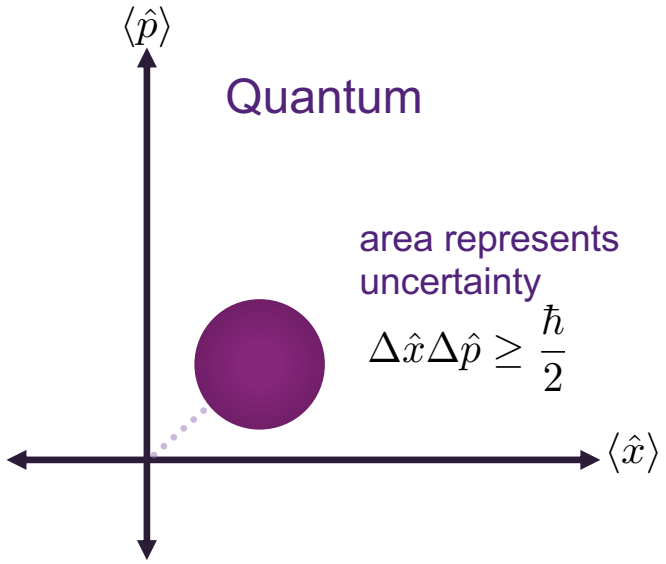
- a centre in phase space;
- an uncertainty in each direction.

The centre is

$$(\langle \hat{x} \rangle, \langle \hat{p} \rangle)$$

with widths Δx^2 and Δp^2.





Coherent state

Expectation value of $\hat{X}_1$

We use $|\alpha\rangle = \hat{D}(\alpha)|0\rangle$ and $\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha$

The first quadrature is

$$\hat{X}_1 = \hat{a}^\dagger + \hat{a}$$

Begin with its expectation value:

$$\langle \hat{X}_1 \rangle_\alpha = \langle \alpha | \hat{X}_1 | \alpha \rangle$$

Substitute the displaced vacuum:

$$\langle \hat{X}_1 \rangle_\alpha = \langle 0 | \hat{D}^\dagger(\alpha) \hat{X}_1 \hat{D}(\alpha) | 0 \rangle$$

Substitute the definition of $\hat{X}_1$:

$$\langle \hat{X}_1 \rangle_\alpha = \langle 0 | \hat{D}^\dagger(\alpha) \left(\hat{a}^\dagger + \hat{a}\right) \hat{D}(\alpha) | 0 \rangle$$

Apply the displacement identities:

$$\langle \hat{X}_1 \rangle_\alpha = \langle 0 | \left(\hat{a}^\dagger + \alpha^* + \hat{a} + \alpha\right) | 0 \rangle$$

Coherent state

Expectation value of $\hat{X}_1$

Apply the displacement identities: $$\langle \hat{X}_1 \rangle_\alpha = \langle 0 | \left(\hat{a}^\dagger + \alpha^* + \hat{a} + \alpha \right) | 0 \rangle$$

Separate the terms: $$\langle \hat{X}_1 \rangle_\alpha = \langle 0 | \hat{a}^\dagger | 0 \rangle + \alpha^* + \langle 0 | \hat{a} | 0 \rangle + \alpha$$

The vacuum expectation values vanish:

$$\langle 0 | \hat{a} | 0 \rangle = \langle 0 | \hat{a}^\dagger | 0 \rangle = 0$$

Therefore,

$$\langle \hat{X}_1 \rangle_\alpha = \alpha + \alpha^* = 2\Re(\alpha)$$

Coherent state

Expectation value of $\hat{X}_2$

The second quadrature is $$\hat{X}_2 = i\left(\hat{a}^\dagger - \hat{a}\right)$$

Begin with its expectation value:

$$\langle \hat{X}_2 \rangle_\alpha = \langle \alpha | \hat{X}_2 | \alpha \rangle$$

Substitute the displaced vacuum:

$$\langle \hat{X}_2 \rangle_\alpha = \langle 0 | \hat{D}^\dagger(\alpha) \hat{X}_2 \hat{D}(\alpha) | 0 \rangle$$

Substitute the definition of $\hat{X}_2$:

$$\langle \hat{X}_2 \rangle_\alpha = i\langle 0 | \hat{D}^\dagger(\alpha) \left(\hat{a}^\dagger - \hat{a}\right) \hat{D}(\alpha) | 0 \rangle$$

Apply the displacement identities:

$$\langle \hat{X}_2 \rangle_\alpha = i\langle 0 | \left(\hat{a}^\dagger + \alpha^* - \hat{a} - \alpha\right) | 0 \rangle$$

Coherent state

Expectation value of $\hat{X}_2$

Apply the displacement identities:

$$\langle \hat{X}_2 \rangle_\alpha = i\langle 0| \left(\hat{a}^\dagger + \alpha^* - \hat{a} - \alpha\right) |0\rangle$$

Separate the terms:

$$\langle \hat{X}_2 \rangle_\alpha = i\left(\langle 0|\hat{a}^\dagger|0\rangle + \alpha^* - \langle 0|\hat{a}|0\rangle - \alpha\right)$$

The vacuum expectation values vanish:

$$\langle \hat{X}_2 \rangle_\alpha = i\left(\alpha^* - \alpha\right)$$

Using the imaginary part of α, $\alpha^* - \alpha = -2i\Im(\alpha)$

Therefore, $\langle \hat{X}_2 \rangle_\alpha = 2\Im(\alpha)$

Coherent state

Variance of $\hat{X}_1$

The variance is defined by

$$\mathrm{var}_\alpha(\hat{X}_1) = \langle \hat{X}_1^2 \rangle_\alpha - \langle \hat{X}_1 \rangle_\alpha^2$$

$$\langle \hat{X}_1^2 \rangle_\alpha = \langle \alpha | \hat{X}_1^2 | \alpha \rangle$$

Write the coherent state as a displaced vacuum:

$$\langle \hat{X}_1^2 \rangle_\alpha = \langle 0 | \hat{D}^\dagger(\alpha) \hat{X}_1^2 \hat{D}(\alpha) | 0 \rangle$$

Transform the quadrature operator:

$$\hat{D}^\dagger(\alpha) \hat{X}_1 \hat{D}(\alpha) = \hat{X}_1 + 2\Re(\alpha)$$

Therefore,

$$\langle \hat{X}_1^2 \rangle_\alpha = \langle 0 | \left(\hat{X}_1 + 2\Re(\alpha) \right)^2 | 0 \rangle$$

Coherent state

Variance of $\hat{X}_1$

$$\langle \hat{X}_1^2 \rangle_\alpha = \langle 0 | \left(\hat{X}_1 + 2\Re(\alpha) \right)^2 |0\rangle$$

Expand the square: $\langle \hat{X}_1^2 \rangle_\alpha = \langle 0 | \hat{X}_1^2 | 0 \rangle + 4\Re(\alpha) \langle 0 | \hat{X}_1 | 0 \rangle + 4\Re(\alpha)^2$

For the vacuum, $\langle 0 | \hat{X}_1 | 0 \rangle = 0$

Also, $\langle 0 | \hat{X}_1^2 | 0 \rangle = 1$

Thus, $\langle \hat{X}_1^2 \rangle_\alpha = 1 + 4\Re(\alpha)^2$

Coherent state

Variance of $\hat{X}_1$

$$\langle \hat{X}_1^2 \rangle_\alpha = 1 + 4\Re(\alpha)^2$$

We already found

$$\langle \hat{X}_1 \rangle_\alpha^2 = 4\Re(\alpha)^2$$

Therefore,

$$\mathrm{var}_\alpha(\hat{X}_1) = 1 + 4\Re(\alpha)^2 - 4\Re(\alpha)^2$$

Finally,

$$\boxed{\mathrm{var}_\alpha(\hat{X}_1) = 1}$$

Generalised quadrature operator

A quadrature at an arbitrary phase-space angle θ is defined by

$$\hat{X}_\theta = \hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta}$$

The operator is Hermitian: $\hat{X}_\theta^\dagger = \hat{X}_\theta$

It can also be written as a rotated combination of the original quadratures:

$$\hat{X}_\theta = \hat{X}_1 \cos\theta + \hat{X}_2 \sin\theta$$

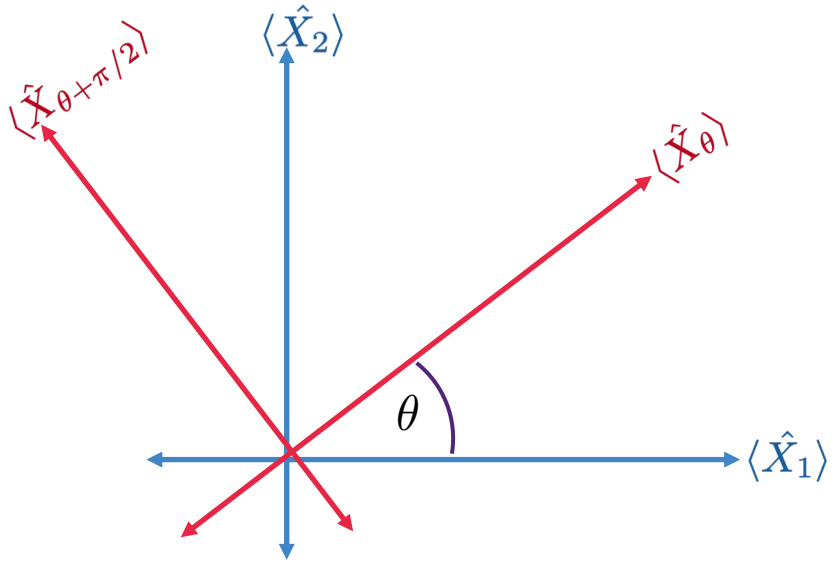
The original quadratures correspond to

$$\hat{X}_{\theta=0} = \hat{X}_1, \qquad \hat{X}_{\theta=\pi/2} = \hat{X}_2$$

The quadrature orthogonal to $\hat{X}_\theta$ is $\hat{X}_{\theta+\pi/2}$

The two orthogonal quadratures satisfy

$$\left[\hat{X}_\theta, \hat{X}_{\theta+\pi/2}\right] = 2i$$



The coherent-state variance is independent of the angle:

$$\boxed{\mathrm{var}_\alpha(\hat{X}_\theta) = 1}$$

Coherent state

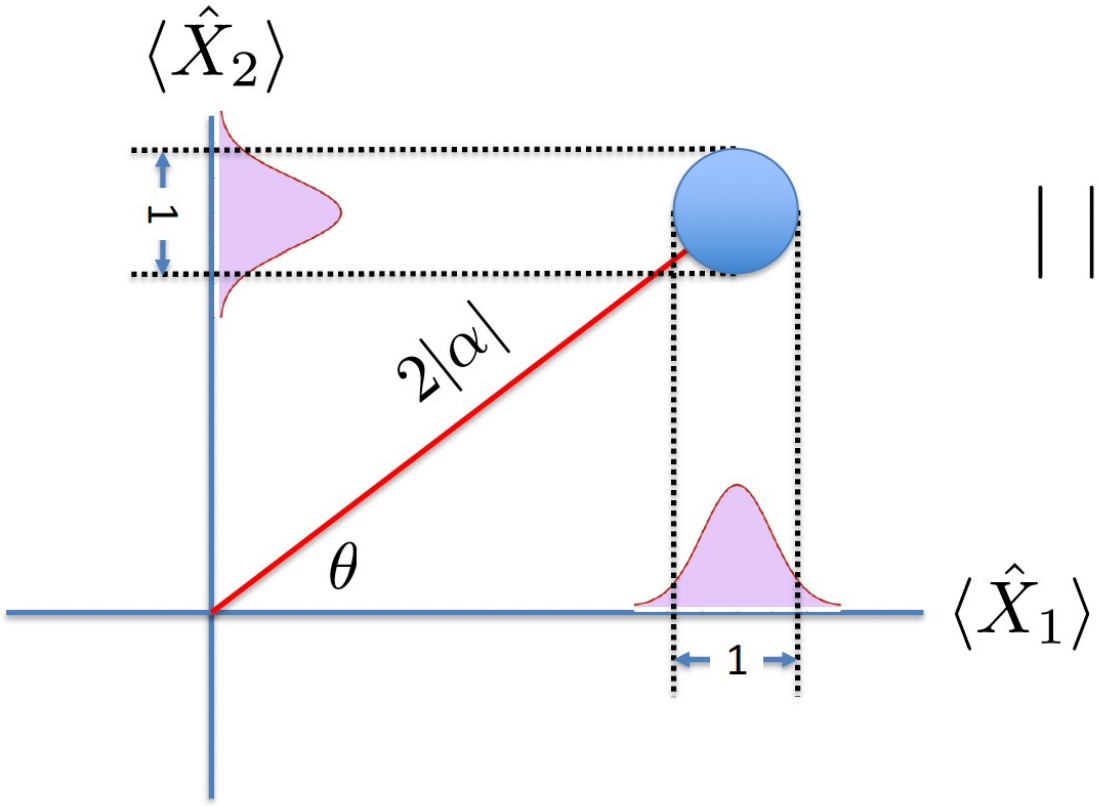
$$\langle \hat{X}_1 \rangle_\alpha = 2\Re(\alpha)$$

$$\langle \hat{X}_2 \rangle_\alpha = 2\Im(\alpha)$$

$$\boxed{\mathrm{var}_\alpha(\hat{X}_1) = 1}$$

$$\boxed{\mathrm{var}_\alpha(\hat{X}_2) = 1}$$

$$\boxed{\mathrm{var}_\alpha(\hat{X}_\theta) = 1}$$



$$|\,|\alpha|e^{-i\theta}\rangle$$

Vacuum fluctuations

For the vacuum state,

$$\langle \hat{X}_1 \rangle = \langle 0 | \hat{a} + \hat{a}^\dagger | 0 \rangle = 0, \qquad \langle \hat{X}_2 \rangle = 0$$

The second moments are

$$\langle \hat{X}_1^2 \rangle = 1, \qquad \langle \hat{X}_2^2 \rangle = 1$$

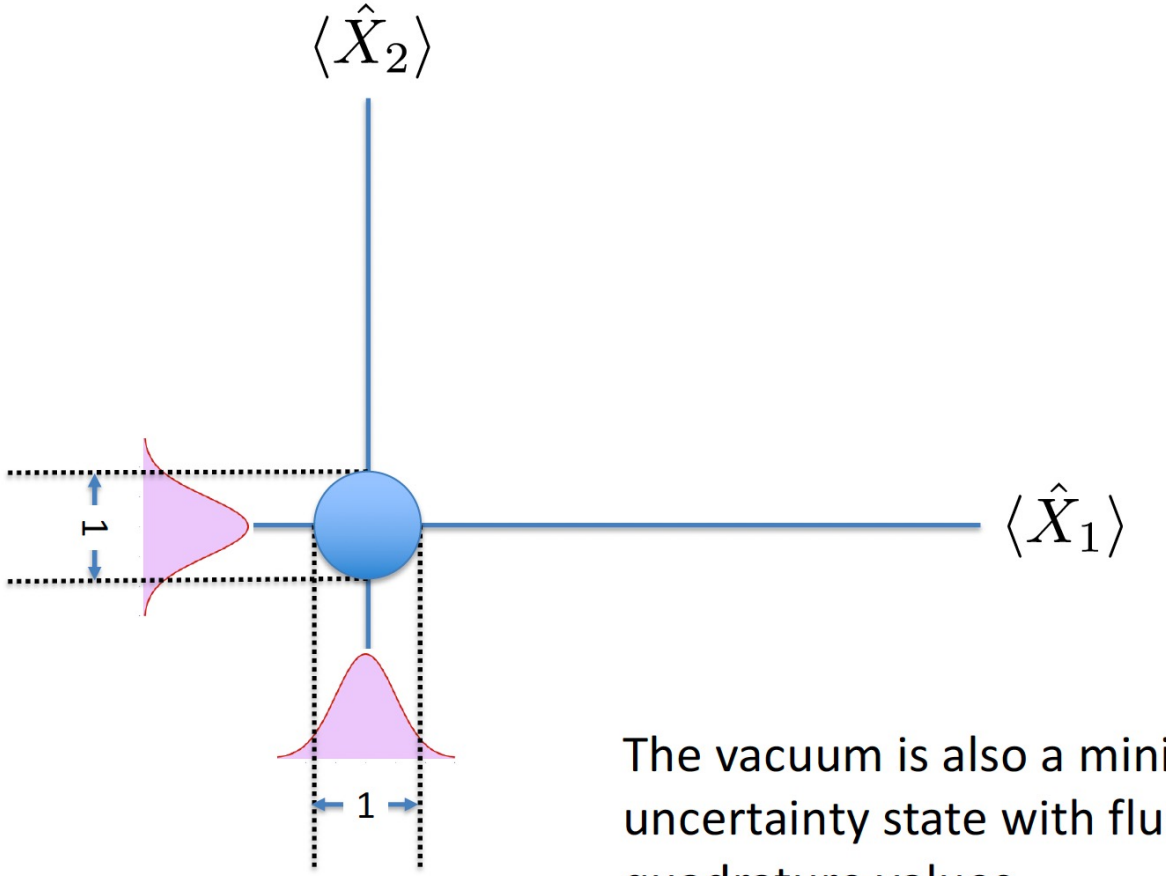
Therefore,

$$\mathrm{var}(\hat{X}_1) = \mathrm{var}(\hat{X}_2) = 1$$

The vacuum saturates the uncertainty relation:

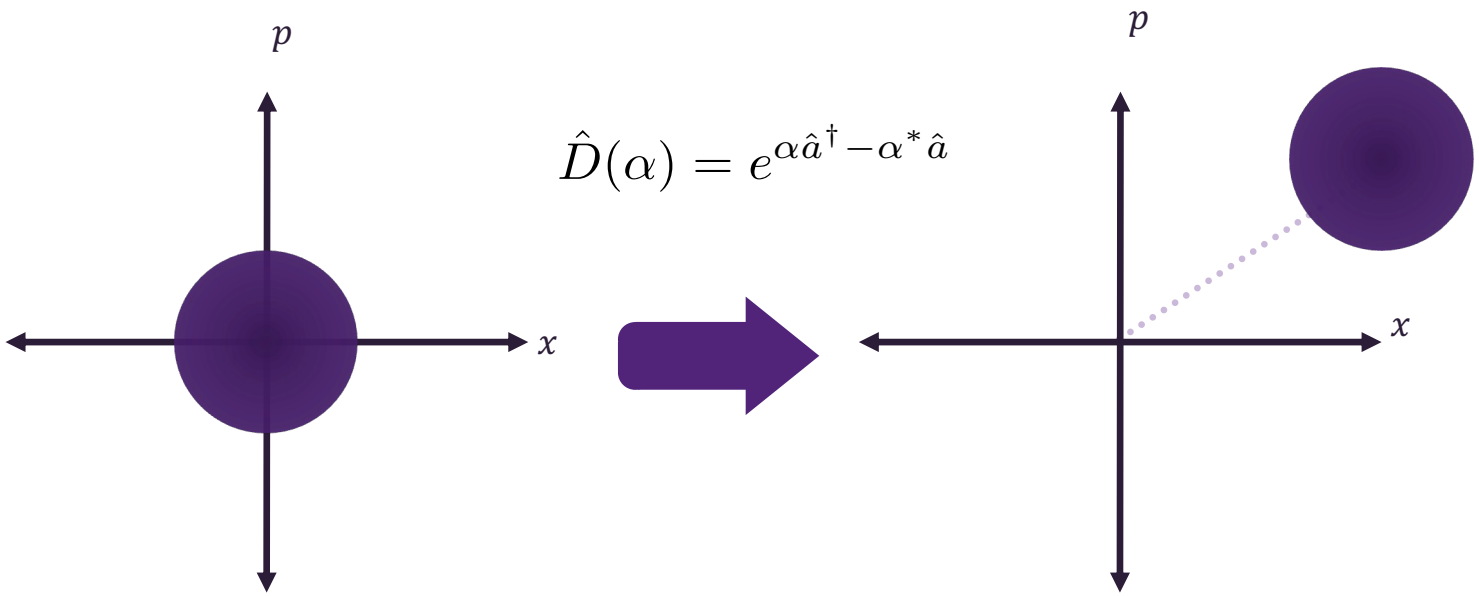
$$\Delta X_1 \Delta X_2 = 1$$

Vacuum fluctuations



The vacuum is also a minimum uncertainty state with fluctuating quadrature values.

Displacing the vacuum in phase space



Energy of a coherent state

For a single optical mode, the Hamiltonian is $\hat{H} = \frac{1}{2}\hbar\Omega\left(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right)$

Using the quadrature operators, $\hat{H} = \frac{1}{4}\hbar\Omega\left(\hat{X}_1^2 + \hat{X}_2^2\right)$

The expectation value in a coherent state is $\langle\hat{H}\rangle_\alpha = \frac{1}{4}\hbar\Omega\left(\langle\hat{X}_1^2\rangle_\alpha + \langle\hat{X}_2^2\rangle_\alpha\right)$

From the previous calculations, $\langle\hat{X}_1^2\rangle_\alpha = 1 + 4\Re(\alpha)^2$ and $\langle\hat{X}_2^2\rangle_\alpha = 1 + 4\Im(\alpha)^2$

Substituting these results gives $\langle\hat{H}\rangle_\alpha = \frac{1}{4}\hbar\Omega\left[2 + 4\Re(\alpha)^2 + 4\Im(\alpha)^2\right]$

Using $|\alpha|^2 = \Re(\alpha)^2 + \Im(\alpha)^2$

we obtain $\boxed{\langle\hat{H}\rangle_\alpha = \hbar\Omega\left(|\alpha|^2 + \frac{1}{2}\right)}$

we have added photons to the vacuum without adding any additional noise.

Time evolution of a coherent state

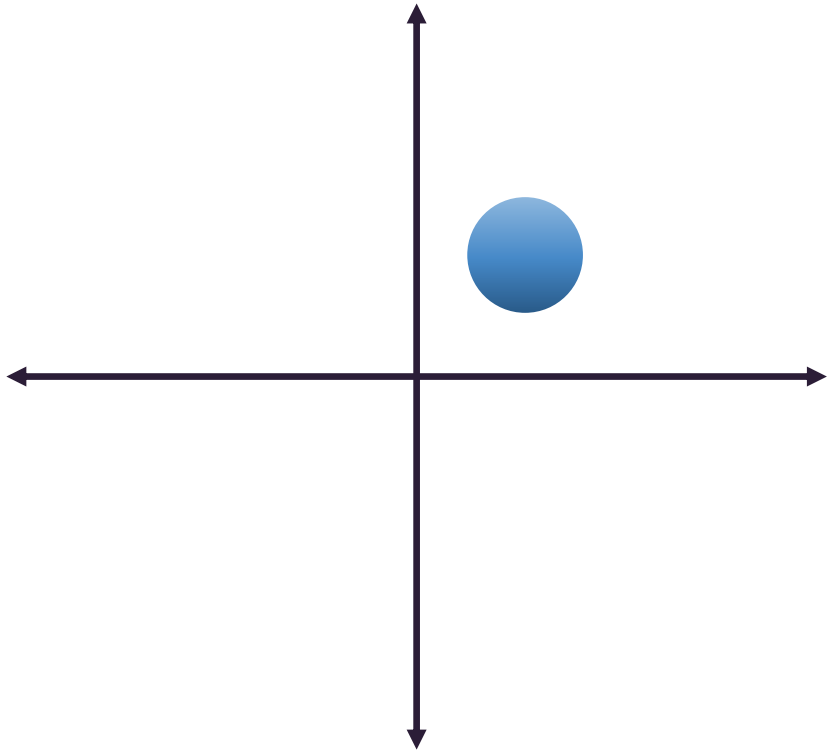
Under free evolution, a coherent state remains coherent:

$$\alpha(t) = \alpha(0)e^{-i\Omega t}$$

The state rotates around the origin of phase space.

Its distance from the origin remains constant.

Its circular uncertainty region does not change shape.

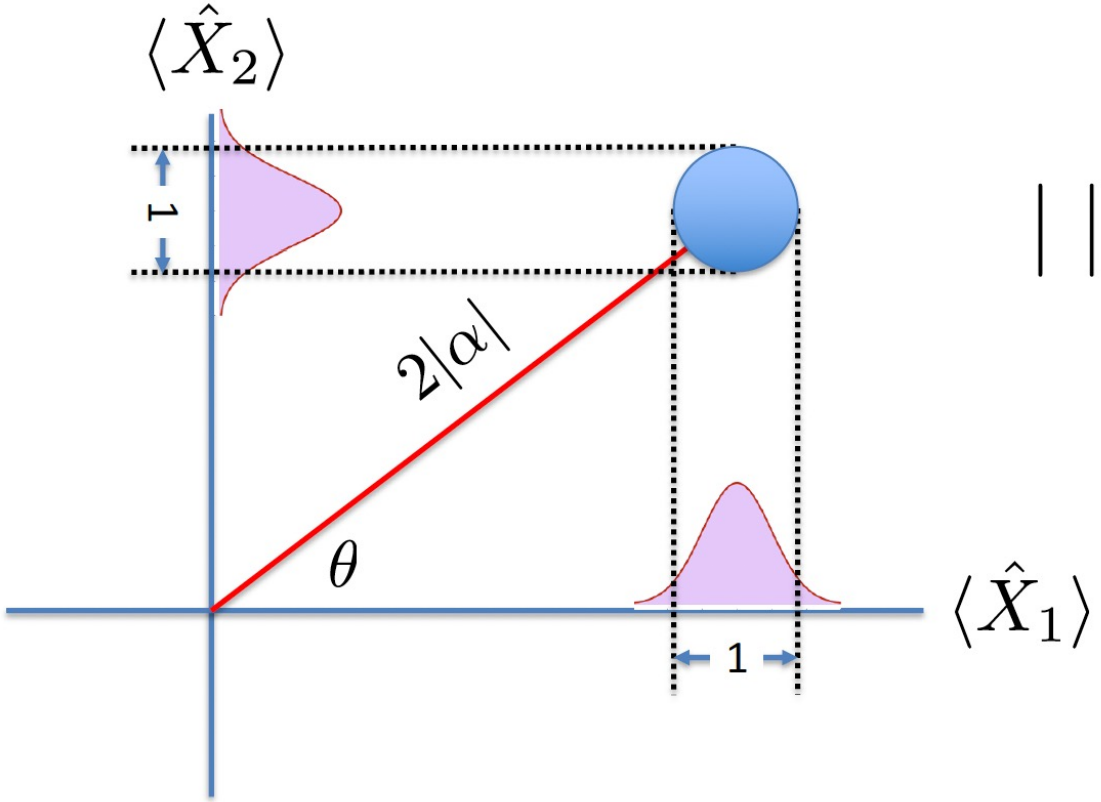


Measurement of a coherent state

The phase is measured relative to a local oscillator using **homodyne detection**.

The measured quadrature is selected by the local-oscillator phase θ:

$$\langle \hat{X}_\theta \rangle_\alpha = 2|\alpha| \cos(\phi - \theta)$$



$$|\,|\alpha|e^{-i\theta}\rangle$$

Number states

- For a one photon number state the expectation value of the X_1 quadrature is:

$$\langle \hat{X}_1 \rangle = \langle 1 | (\hat{a}^\dagger + \hat{a}) | 1 \rangle = 0$$

- And the expectation value of the X_2 quadrature is:

$$\langle \hat{X}_2 \rangle = \langle 1 | i(\hat{a}^\dagger - \hat{a}) | 1 \rangle = 0$$

- And of course the photon number is:

$$\langle \hat{n} \rangle = \langle 1 | \hat{a}^\dagger \hat{a} | 1 \rangle = 1$$

Number states

- For a one photon number state the variance of the X_1 quadrature is:

$$\Delta\hat{X}_1^2 = \langle\hat{X}_1^2\rangle - \langle\hat{X}_1\rangle^2 = 3$$

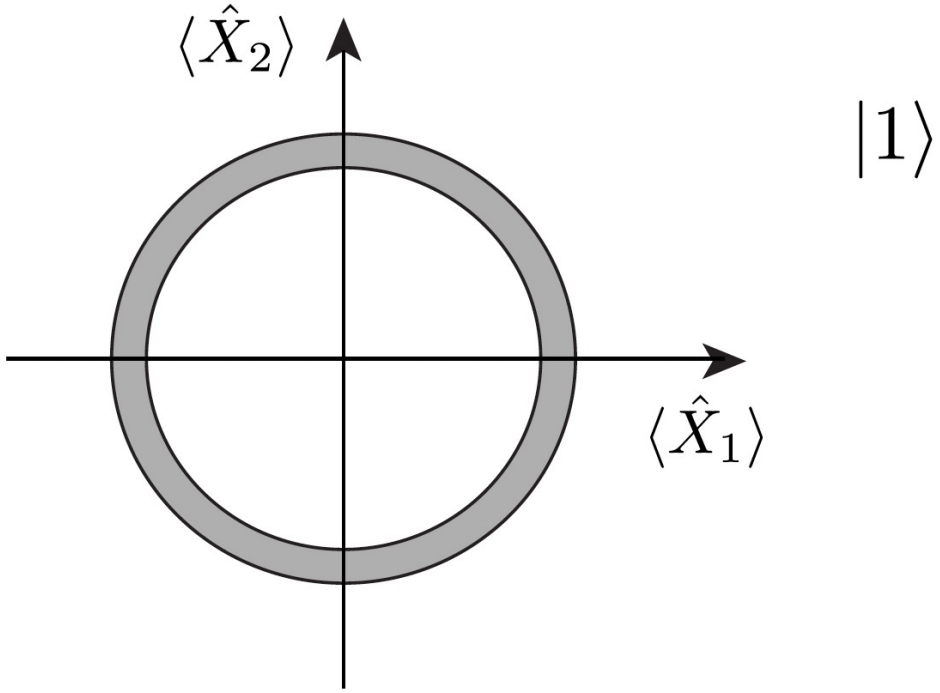
- And the expectation value of the X_2 quadrature is:

$$\Delta\hat{X}_2^2 = \langle\hat{X}_2^2\rangle - \langle\hat{X}_2\rangle^2 = 3$$

- So number states are not minimum uncertainty states:

$$\Delta\hat{X}_1^2\Delta\hat{X}_2^2 = 9 > 1$$

Number states



$|1\rangle$

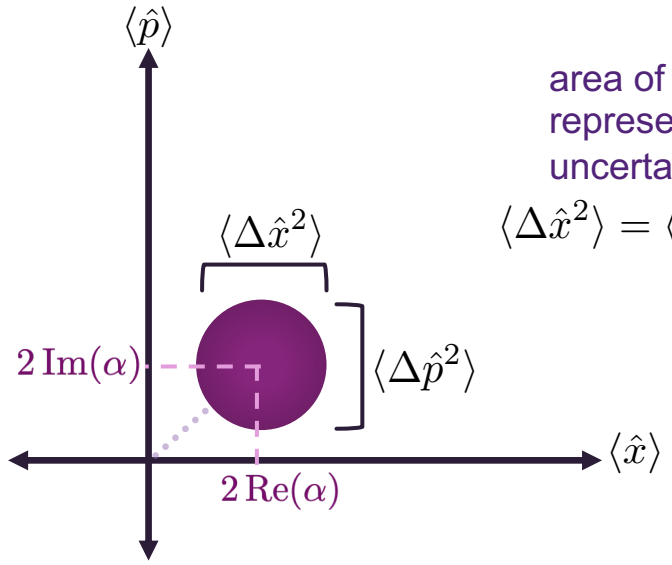
The phase space profile of a single photon is a ring!

Summary

Quadrature operators can be used to visualise states on a two-dimensional axis

$$\hat{x} = \left(\hat{a}^\dagger + \hat{a}\right)$$
$$\hat{p} = i\left(\hat{a}^\dagger - \hat{a}\right)$$

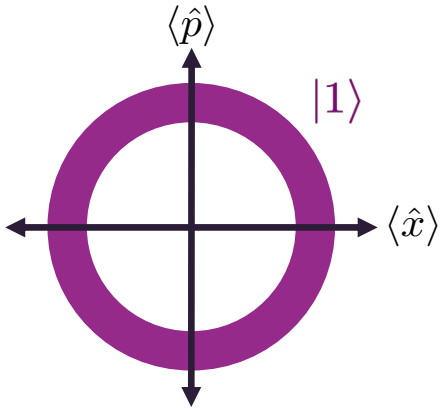
X and P can be measured by homodyne detection



area of circle represents uncertainty

$$\langle\Delta\hat{x}^2\rangle = \langle\Delta\hat{p}^2\rangle = 1$$

Coherent state



Number state